Figure 3c and 3d of the Dunkelflaute paper: how they are made and how robust they are

Internal note for Jan Streffing and Thomas Jung

10 November 2026

1 Summary

The question was whether the geostrophic speed G really weakens this clearly within the weather regimes. It does. Single winters are noisy, but over decades the weakening under Scandinavian blocking (ScBL) stands out in every run and in nearly every CMIP6 model. We rebuilt Fig. 3c and 3d from the daily data, reproduced the stored numbers to machine precision, and tested both panels against a bootstrap over winters and against unforced variability in the 147-winter constant-forcing control of the 100 km run.

- The within-regime term of Fig. 3c lies outside every placement of the control null in all four runs. The regime-frequency term is small and cannot be told apart from zero.
- Under ScBL, G weakens beyond the null in all four runs and beyond the winter-to-winter variability in 18 of 20 CMIP6 models, and all 20 agree on the sign.
- The weakening sits in the weather systems, whose pressure gradients fall at nearly unchanged amplitude. The mean pressure field does not cause it, since its gradient over the sector strengthens.
- We found no earlier study with this result for wind droughts.

2 From daily pressure to regimes

The input is daily-mean sea-level pressure (msl) on a 2.5° grid over 20–80°N and 80°W–40°E, each value the mean of all native points within the grid box. The regimes are defined on ERA5 alone. We removed a mean and three annual harmonics from the 9976 September-to-March days of 1979/80–2024/25, computed EOFs, and clustered the first 10 principal components, which explain 82 % of the variance, into four regimes with k-means. Model days are not clustered. Their anomalies from the seasonal cycle of the comparison's control are projected on the ERA5 EOFs, and each day goes to the nearest ERA5 centroid.

The ERA5 patterns are a fair basis for the models (Fig. 1). The first three model EOFs match ERA5 with pattern correlations of 0.87 to 0.99. Clustering each run on its own gives unstable centroids on 10 to 17 winters, as it does for ERA5 sub-periods of the same length, which is why we assign model days to the ERA5 centroids instead. The composites of the assigned days reproduce the ERA5 centroids with $r \geq 0.99$ in the controls and $r \geq 0.98$ in all CMIP6 historical runs.

3 Figure 3c step by step

A low-wind event is a spell in which the 48 h mean of the region's wind capacity factor stays below a threshold. The threshold is set separately in each region and run so that the control has five events per winter, which makes it a percentile of that run's own distribution, and the future is counted at the same threshold. Each event takes the regime that covers most of its days. For one region and period, let f_k be the regime-k days per winter and λ_k the events per regime-k day, so that the events per winter are $e = \sum_k f_k \lambda_k$. The change from control to future splits exactly into

$$\Delta e \;=\; \underbrace{\sum_k \lambda_k^{\mathrm{ctl}} \left(f_k^{\mathrm{fut}} - f_k^{\mathrm{ctl}}\right)}_{\text{frequency term}} \;+\; \underbrace{\sum_k f_k^{\mathrm{fut}} \left(\lambda_k^{\mathrm{fut}} - \lambda_k^{\mathrm{ctl}}\right)}_{\text{within-regime term}} . \tag{1}$$

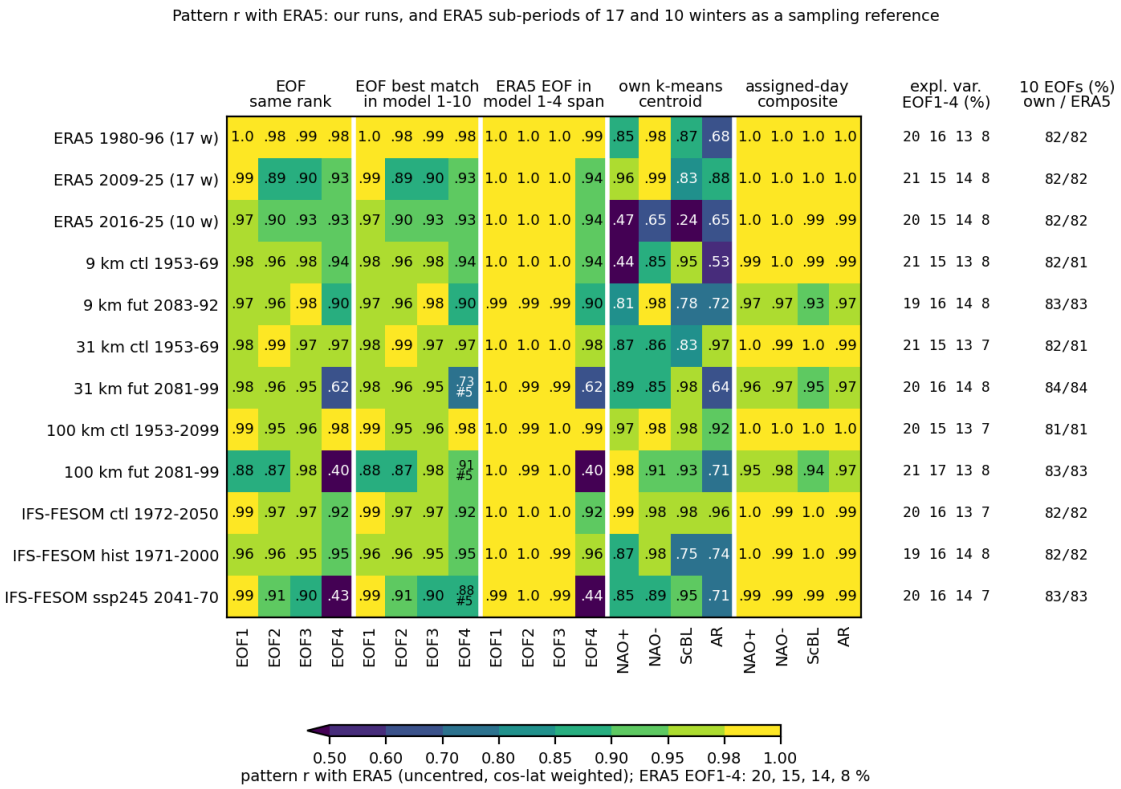


Figure 1: Pattern correlation with ERA5 for our runs and for ERA5 sub-periods of 17 and 10 winters. The paper uses only the last block, the composites of the assigned days.

Both terms are computed per region, averaged over the eight European regions, and divided by the global-mean warming ΔT of the comparison.

Table 1 goes through Germany at 9 km. ScBL produces the most events per regime day, and under warming its rate rises from 4.5 to 6.5 events per 100 days. Its within-regime part is the future number of ScBL days times this rise, about one event per winter. Its frequency part is the control rate times the loss of nine ScBL days, about -0.4. Summed over the four regimes, Germany gains 1.5 events per winter, a within-regime gain of 2.0 of which the frequency term takes back 0.45.

The bootstrap resamples whole winters, independently in control and future and with the same winters in every region. The null places two disjoint blocks with the comparison's numbers of winters in all possible positions within the 147-winter control and treats the second block as the future. The within-regime term exceeds every placement of the null in all four runs (Table 2, Fig. 2). The frequency term is negative in all four runs, and its bootstrap range includes zero in all four. ScBL carries the largest share of the within-regime term in three of four runs and is significant in all four. Splitting tied events between regimes or using the symmetric form of Eq. 1 shifts the terms by at most 0.015 events per winter per K.

Table 1: Worked example, Germany at 9 km, control 1953–1969 against future 2083–2092. f_k in regime days per winter, e_k in events per winter, λ_k in events per 100 regime days, and the two terms of Eq. 1 in events per winter per K.

	f_k		e_k		$100\,\lambda_k$		frequency	within
regime	ctl	fut	ctl	fut	ctl	fut	per K	per K
NAO+	53.4	70.7	0.76	1.60	1.43	2.26	+0.045	+0.107
NAO−	54.8	34.9	1.12	0.40	2.04	1.15	−0.074	−0.057
ScBL	59.8	50.9	2.71	3.30	4.52	6.48	−0.073	+0.181
AR	44.3	55.8	0.41	1.20	0.93	2.15	+0.019	+0.124
sum	212.2	212.3	5.00	6.50			−0.082	+0.355

Table 2: Terms of Fig. 3c, 8-region mean in events per winter per K, with the bootstrap 5–95 % range below each value and the one-sided p of the control null. The last column counts the regions with a positive within-regime term.

			total		frequency term		within-regime term		
run	ΔT (K)	winters	value 5–95 %	p	value 5–95 %	p	value 5–95 %	p	> 0
9 km	5.50	17/10	+0.246 [+0.099, +0.400]	<0.001	−0.036 [−0.074, +0.004]	0.039	+0.282 [+0.155, +0.408]	<0.001	8
31 km	5.68	17/19	+0.233 [+0.123, +0.340]	<0.001	−0.024 [−0.053, +0.008]	0.066	+0.257 [+0.152, +0.353]	<0.001	8
100 km	4.90	147/19	+0.220 [+0.128, +0.309]	<0.008	−0.014 [−0.041, +0.014]	0.155	+0.234 [+0.152, +0.313]	<0.008	8
IFS-FESOM	2.30	79/30	+0.135 [−0.018, +0.290]	0.015	−0.028 [−0.084, +0.028]	0.100	+0.163 [+0.023, +0.302]	<0.001	7

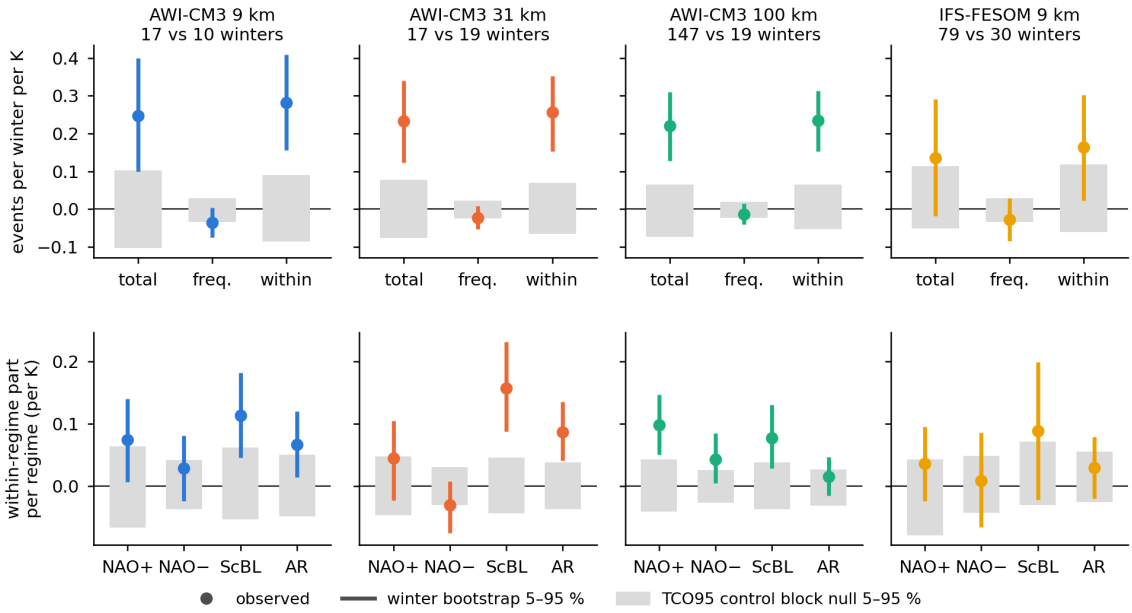


Figure 2: Robustness of Fig. 3c. Top, the total and the two terms of each run. Bottom, the within-regime term split by regime. Dots are the observed values, bars the winter bootstrap 5–95 % range, and grey boxes the 5–95 % range of the block null from the 147-winter 100 km control.

4 Figure 3d step by step

From the daily msl p we computed $G = |\nabla p|/(\rho f)$ with centred differences, $\rho = 1.25\,\mathrm{kg\,m^{-3}}$ and $f = 2\Omega \sin\varphi$, averaged over each region box. For each region and regime we compared the mean G of all regime days in future and control, averaged the change over the eight regions, and expressed it per kelvin as $100\,\big(\exp[\ln(1 + \Delta G/100)/\Delta T] - 1\big)$. Our runs use the msl of the regime analysis, and CMIP6 uses daily psl interpolated bilinearly onto the same 2.5° grid. On the bilinear footing our values change by at most 0.23 % per K.

Under ScBL the weakening exceeds every placement of the null in all four runs (Fig. 3, Table 3). In CMIP6 all 20 models weaken under ScBL, and in 18 the bootstrap range lies below zero. The other regimes weaken less and are significant only in some runs. A single winter is noisy, with a standard deviation of 4 to 8 % in winter-mean ScBL G, so the signal emerges only over decades. Labelling the future with its own seasonal cycle, or keeping only the ScBL days closest to the centroid, leaves the weakening in place.

5 Why the models agree, and whether this is known

We split each day's pressure field into the seasonal cycle of its period and the anomaly, and recomputed G with the two swapped between control and future. The weakening comes almost entirely from the anomalies, the weather, and this anomaly term is negative in every regime, every run and all 20 CMIP6 models (Fig. 4). The mean pressure field does not explain it, since its north–south gradient over the sector strengthens as pressure

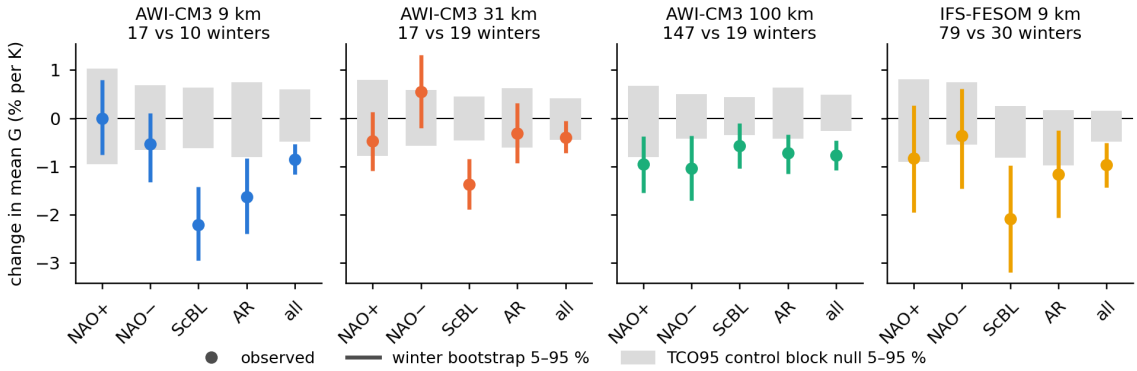


Figure 3: Change in mean G within each regime and over all days, % per K, eight regions. Dots, bootstrap bars and null boxes as in Fig. 2.

Table 3: Change in mean G within regimes, % per K. For our runs the value of Fig. 3d with its bootstrap 5–95 % range, the null p, and the value on the bilinear CMIP6 footing ("bil."). For CMIP6 the median and the 10–90 % range of 20 models, the number of models with a negative change and the number whose bootstrap range lies below zero.

regime	9 km	31 km	100 km	IFS-FESOM	CMIP6
NAO+	−0.01 [−0.76, +0.79] p 0.494; bil. +0.07	−0.48 [−1.09, +0.13] p 0.184; bil. −0.36	−0.95 [−1.54, −0.37] p 0.008; bil. −0.93	−0.83 [−1.95, +0.27] p 0.072; bil. −0.69	−0.66 [−1.52, +0.04] 17 neg.; 9 sig.
NAO−	−0.54 [−1.32, +0.10] p 0.101; bil. −0.51	+0.55 [−0.21, +1.32] p 0.066; bil. +0.57	−1.03 [−1.70, −0.36] p <0.008; bil. −1.04	−0.36 [−1.45, +0.61] p 0.208; bil. −0.27	−0.51 [−1.16, +0.16] 17 neg.; 8 sig.
ScBL	−2.21 [−2.94, −1.42] p <0.001; bil. −2.06	−1.38 [−1.88, −0.84] p <0.001; bil. −1.25	−0.57 [−1.04, −0.11] p <0.008; bil. −0.57	−2.08 [−3.19, −0.97] p <0.001; bil. −1.87	−1.13 [−1.74, −0.62] 20 neg.; 18 sig.
AR	−1.62 [−2.40, −0.83] p <0.001; bil. −1.40	−0.31 [−0.93, +0.31] p 0.200; bil. −0.24	−0.72 [−1.14, −0.34] p <0.008; bil. −0.71	−1.16 [−2.06, −0.25] p 0.002; bil. −1.01	−1.14 [−1.67, −0.25] 19 neg.; 14 sig.
all	−0.85 [−1.16, −0.53] p 0.001; bil. −0.75	−0.39 [−0.72, −0.06] p 0.067; bil. −0.31	−0.77 [−1.08, −0.46] p <0.008; bil. −0.76	−0.96 [−1.43, −0.51] p <0.001; bil. −0.82	−0.89 [−1.23, −0.58] 20 neg.; 19 sig.

falls over the Arctic and Nordic seas. Under ScBL this background term is close to zero, whereas in the other regimes it is positive and offsets part of the weakening, which is why blocking shows the clearest net signal. The blocking high itself hardly changes in peak, extent or position. What weakens is the gradient of the day-to-day deviations from the regime pattern, at nearly unchanged amplitude, so the synoptic disturbances become broader and smoother.

We found no study that splits projected changes in wind droughts into a regime-frequency and a within-regime part, and none that shows weaker pressure gradients within blocking across models and per kelvin of warming. For precipitation, the change of the weather within fixed regimes outweighs the change of

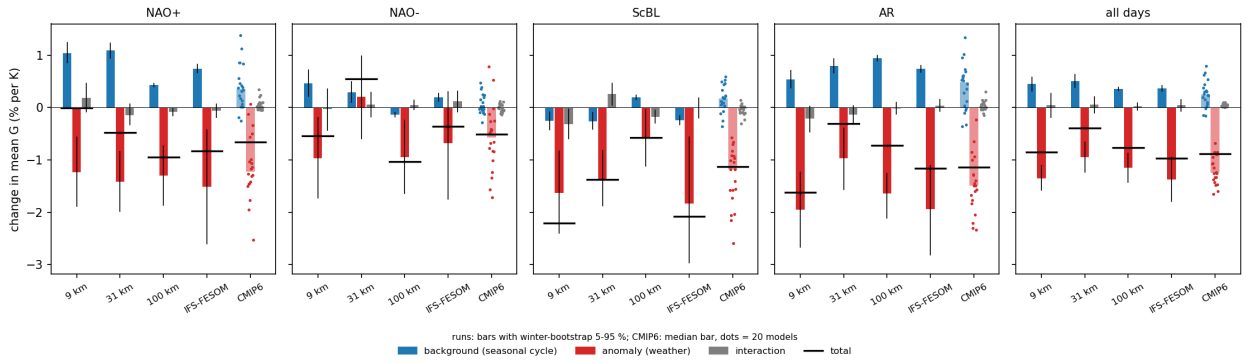


Figure 4: Change of mean G per regime split into a part from the seasonal cycle (background, blue), from the anomalies (weather, red), and their interaction (grey). Black bars are the totals of Fig. 3d. Our runs with winter-bootstrap 5–95 % ranges, CMIP6 as median bars with one dot per model.

their frequency [1, 2]. Broader disturbances match the projected increase of the eddy length scale under warming [4]. A weaker low-level temperature gradient between equator and pole damps synoptic anomalies in idealised models [3, 6] and tracks projected European offshore wind [7], and the geostrophic wind from sea-level pressure declines over the Northeast Atlantic in 32 CMIP6 models [5]. Neither the mean pressure gradient nor a regional temperature contrast explains the spread between our models ($|r| \leq 0.26$), so the weakening is a robust change of the weather systems whose cause remains open.

6 Changes made in the paper

The paper now states the robust result plainly and leaves the tests to the Supplement. The change in events is almost entirely a change within regimes, well beyond unforced variability, and under ScBL G falls in all 20 CMIP6 models and in all four of our runs. The Discussion says that the weakening sits in the weather systems and not in the mean state, consistent with broader eddies, and that its cause remains open.

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